American Computer Science League

2024-2025 ● Contest 1: Short Problems Solutions ● Intermediate Division

1. Computer Number Systems

Convert each hex digit to binary.
$2024_{16} = 0010\ 0000\ 0010\ 0100_2$
Now drop the leading zeros. 10000000100100

B. 10000000100100

2. Computer Number Systems

$$\begin{array}{r} 111 \\ 4265_8 \\ +\ 6524_8 \\ \hline 13011_8 \end{array}$$

D. 13011

3. Recursive Functions

$f(1) = 256$
$f(2) = f(2-1)/2 = f(1)/2 = 256/2 = 128$
$f(3) = f(3-1)/2 = f(2)/2 = 128/2 = 64$
$f(4) = f(4-1)/2 = f(3)/2 = 64/2 = 32$
$f(5) = f(5-1)/2 = f(4)/2 = 32/2 = 16$

E. 16

4. Recursive Functions

$f(12,10) = f(12-3,10+1) - 2 = f(9,11) - 2 = 8 - 2 = 6$
$f(9,11) = f(9+1,11-4) + 1 = f(10,7) + 1 = 7 + 1 = 8$
$f(10,7) = f(10-3,7+1) - 2 = f(7,8) - 2 = 9 - 2 = 7$
$f(7,8) = f(7+1,8-4+1 = f(8,4) + 1 = 8 + 1 = 9$
$f(8,4) = f(8-3,4+1) - 2 = f(5,5) - 2 = 10 - 2 = 8$
$f(5,5) = 2 * 5 = 10$ Now substitute backwards.

A. 6

5. What Does This Program Do?

In tracing the following program, integer division gets the quotient and the operator % (mod) gets the remainder.

`h = 200`
`y = int(h / 36)`
`h = h % 36`
`f = int(h / 12)`
`i = h % 12`
`output y, f, i`
	This program converts a height in total inches to the height in yards, feet, and inches since 36 inches = 1 yard and 12 inches = 1 foot.
	y = int(200/36) = 5 yards
	h = 200 - 5 * 36 = 20
	f = int(20 / 12) = 1 foot
	i = 20 % 12 = 8 inches
	Therefore, output 5,1,8.

C. 5, 1, 8

6. What Does This Program Do?

In tracing the following program, the final value of s is 40 as follows:

`s = 0`
`for x = 1 to 5`
`  s = s + x ^ 2`
`next x`
`for y = 2 to 10 step 2`
`  s = s + y / 2`
`next y`
`for z = 5 to 1 step -1`
`  s = s - z * 2`
`next z`
`output s`
	In the first loop
	s = 1 + 4 + 9 + 16 + 25 = 5 + 25 + 25 = 55.

	In the second loop s = 55 + 1 + 2 + 3 + 4 + 5 = 70.
	In the third loop s = 70 - 10 - 8 - 6 - 4 - 2 = 40.
	Therefore, s = 40.

A. 40